

AMNA FIAZ

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PROFESSIONAL SUMMARY

AI/ML Engineer building production AI systems, agentic RAG assistants, document-verification platforms, and data pipelines on GCP and AWS. Cut end-to-end data-processing time by 40 per cent across 8 clinical datasets, and built collection pipelines handling 50,000+ records per run. Redis Associate Developer certified. Worked on projects involving LLMs, RAG, OCR, PDF verification, web automation, data engineering, and multi-agent workflows, with a strong focus on building reliable systems that solve real business problems.

TECHNICAL SKILLS

AI & LLM Systems	RAG system, LangChain, LangGraph workflow, LLM Agents, Tool Calling, FAISS, ChromaDB, Groq, HuggingFace, Prompt Engineering, LLM Evaluation
Python & Backend	Python, FastAPI, Django, REST APIs, GraphQL, Pydantic, SQLAlchemy, Uvicorn, async/await, Celery
Data & Databases	PostgreSQL, MySQL, BigQuery, SQL, Pandas, NumPy, schema design, query optimization, data validation
Document AI & Automation	PyMuPDF, PaddleOCR, Playwright, PDF Parsing, OCR, Regex Extraction, Template Matching, n8n
Cloud & DevOps	Docker, Docker Compose, Git, GitHub Actions, Linux, AWS EC2, GCP, Cloud Composer, Redis, SSH

WORK EXPERIENCE

Black Lion Innovations Group — Data Scientist Mar 2026 – Present

- Built and deployed FastAPI services and automated data pipelines on GCP and AWS EC2, using Docker, PostgreSQL, BigQuery, background jobs, and production monitoring.
- Built an AI assistant for women's hormonal health using RAG and agent-based workflows, allowing the system to search trusted knowledge, keep conversation context, and give more relevant answers to users.
- Developed Playwright-based collection pipelines for Spotify artist, playlist, and web data, processing more than 50,000 records in a run with background workers and isolated subprocesses.
- Added automated checks for missing values, duplicates, incorrect schemas, and inconsistent records across more than 15 data sources, cutting down manual data-review work.

LUMS BIRL — AI Engineer May 2025 – Feb 2026

- Implemented Python ETL pipelines and FastAPI services for cleaning, transforming, combining, and reporting data from 8 clinical datasets.
- Designed SQL, Pandas, and NumPy workflows for feature engineering, data validation, and anomaly checks used by downstream research teams.
- Improved PostgreSQL queries and processing logic, reducing end-to-end data-processing time by around 40%.
- Parallelized independent processing steps and simplified pipeline logic so larger datasets could be handled more reliably.

Independent Contractor — **LUMS (Prof. Abdul Basit)** — Automation Engineer Project-based

- Built an automated document-verification platform for Spanish energy-efficiency audit claims, covering 11 document types and roughly 40–50 fields per document.
- Developed PDF parsing, OCR, field extraction, validation, and cross-document checks to compare names, IDs, addresses, prices, measurements, signatures, and audit data.
- Generated marked-up PDFs with green, red, and amber boxes directly on the original pages so reviewers could quickly see correct, incorrect, and unverified information.

SKYCODE — Database Developer Intern May 2024 – Aug 2024

- Designed and optimized MySQL and PostgreSQL databases, improving data retrieval speed by approximately 20%.
- Supported database backup, recovery, performance monitoring, and routine database maintenance for application workloads.
- Assisted with backend integration and SQL query optimization for database-driven web applications.
- Helped maintain database reliability by reviewing queries, monitoring performance issues, and supporting data-related troubleshooting.

PROJECTS

Automated Document Verification Platform – Spanish Energy-Efficiency Audits

- Built a document-checking platform for Spanish energy-efficiency claims that reads 11 different document types and checks around 40–50 fields from each document against the recorded audit information.
- Combined PyMuPDF, PaddleOCR, PostgreSQL, and rule-based validation to check DNI/NIE/CIF numbers, cadastral references, addresses, measurements, VAT calculations, prices, signatures, and values repeated across several documents.
- Created annotated PDFs that place green, red, and amber boxes over the exact fields on the page, with explanations showing reviewers what matched, what was wrong, and what still needed manual checking.

Coder Vista – AI-Powered Career Transition Platform

- Configured an AI-powered career transition platform that analyzes a user's CV, identifies transferable and missing skills, and compares them with the requirements of a selected target role.
- Developed skill-gap prioritization using role relevance and job-market demand, helping users understand which missing skills matter most and what they should learn first.
- Generated personalized learning roadmaps with recommended courses, projects, resources, and next steps, supported by career data and real job-market evidence.

ELSA – Women's Hormonal Health Agentic RAG Platform

- Built ELSA, a personalized AI assistant for women's hormonal health that uses symptoms, health history, lifestyle, and cycle-related context to provide evidence-based guidance.
- Developed a context-aware LLM and LangGraph workflow that retrieves relevant medical knowledge, tracks conversation context, and supports structured guidance for areas such as PMS, PCOS, mood, sleep, and thyroid-related symptoms.
- Added safety-focused response controls and privacy-aware backend workflows to make the assistant more reliable for repeated health conversations while reducing generic or unsupported recommendations.

Spotify Data Intelligence & Analytics Platform

- Implemented complete Spotify data platform using FastAPI, Playwright, Django, and GraphQL to collect artist, playlist, and web data and make it available for analytics and reporting.
- Developed secure background data workflows with Pydantic validation, API-key protection, subprocess jobs, Celery, and Redis, with endpoints for status, logs, cancellation, health checks, and downloads.
- Stored and processed structured data with PostgreSQL, with CSV export and optional BigQuery integration, and built analytics views for exploring Spotify performance data.

WhatsApp Visa Processing AI Agent

- Built an automated visa-processing workflow using n8n and the Twilio WhatsApp API so users could send documents and information through WhatsApp.
- Used LlamaParse and Groq to read uploaded documents, extract important information, and pass structured results through the workflow.
- Connected Google Sheets, webhooks, and human approval steps so staff could review important cases before the system continued.

CERTIFICATIONS

- **Redis Associate Developer** — Redis | Aug 2026 – Aug 2028 | [View Credential](#)
- **Agentic Systems with LangGraph** — DataCamp | 2026 | [View Credential](#)
- **Certificate: 3rd International AI Championship** — AIEF/LUMS | Nov 2025 | [View Certificate](#)
- **Joining Data with pandas** — DataCamp | Jul 2026 | [View Credential](#)
- **Amazon Elastic Container Service (AWS ECS)** — KodeKloud | Aug 2026 | [View Credential](#)
- **Model Context Protocol: Advanced Topics** — Anthropic | Aug 2026 | [View Credential](#)

EDUCATION

UET Lahore — BSc Computer Engineering

Sep 2021 – Mar 2025

CGPA: 3.58/4.0